1) a)

$\mathbf{i} \times \mathbf{i} = \mathbf{0}$

1) b)

$\mathbf{i} \times \mathbf{j} = \mathbf{k}$

1) c)

$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$

1) d)

$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$

1) e)

$\mathbf{j} \times \mathbf{j} = \mathbf{0}$

1) f)

$\mathbf{j} \times \mathbf{k} = \mathbf{i}$

1) g)

$\mathbf{k} \times \mathbf{i} = \mathbf{j}$

1) h)

$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$

1) i)

$\mathbf{k} \times \mathbf{k} = \mathbf{0}$

2) a)

$\mathbf{i} \times (\mathbf{j} + \mathbf{k}) = \mathbf{i} \times \mathbf{j} + \mathbf{i} \times \mathbf{k} = \mathbf{k} - \mathbf{j}$

2) b)

$\mathbf{j} \times (\mathbf{i} + \mathbf{k}) = \mathbf{j} \times \mathbf{i} + \mathbf{j} \times \mathbf{k} = -\mathbf{k} + \mathbf{i}$

2) c)

$\mathbf{k} \times (\mathbf{i} + \mathbf{j}) = \mathbf{k} \times \mathbf{i} + \mathbf{k} \times \mathbf{j} = \mathbf{j} - \mathbf{i}$

2) d)

$(\mathbf{i} - \mathbf{j}) \times (\mathbf{j} - \mathbf{k}) = \mathbf{i} \times \mathbf{j} - \mathbf{i} \times \mathbf{k} - \mathbf{j} \times \mathbf{j} + \mathbf{j} \times \mathbf{k}$
$= \mathbf{k} + \mathbf{j} - \mathbf{0} + \mathbf{i} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

2) e)

$(\mathbf{k} - \mathbf{i}) \times (\mathbf{j} - \mathbf{i}) = \mathbf{k} \times \mathbf{j} - \mathbf{k} \times \mathbf{i} - \mathbf{i} \times \mathbf{j} + \mathbf{i} \times \mathbf{i}$
$= -\mathbf{i} - \mathbf{j} - \mathbf{k} + \mathbf{0} = -\mathbf{i} - \mathbf{j} - \mathbf{k}$

2) f)

$(\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (-\mathbf{i} - \mathbf{j} + \mathbf{k})$
$= -\mathbf{i} \times \mathbf{i} - \mathbf{i} \times \mathbf{j} + \mathbf{i} \times \mathbf{k} - \mathbf{j} \times \mathbf{i} - \mathbf{j} \times \mathbf{j} + \mathbf{j} \times \mathbf{k} - \mathbf{k} \times \mathbf{i} - \mathbf{k} \times \mathbf{j} + \mathbf{k} \times \mathbf{k}$
$= \mathbf{0} - \mathbf{k} - \mathbf{j} + \mathbf{k} + \mathbf{0} + \mathbf{i} - \mathbf{j} + \mathbf{i} + \mathbf{0}$
$= 2\mathbf{i} - 2\mathbf{j}$

2) g)

$(\mathbf{k} - \mathbf{j} - \mathbf{i}) \times (\mathbf{j} - \mathbf{i} - \mathbf{k})$
$= \mathbf{k} \times \mathbf{j} - \mathbf{k} \times \mathbf{i} - \mathbf{k} \times \mathbf{k} - \mathbf{j} \times \mathbf{j} + \mathbf{j} \times \mathbf{i} + \mathbf{j} \times \mathbf{k} - \mathbf{i} \times \mathbf{j} + \mathbf{i} \times \mathbf{i} + \mathbf{i} \times \mathbf{k}$
$= -\mathbf{i} - \mathbf{j} - \mathbf{0} - \mathbf{0} - \mathbf{k} + \mathbf{i} - \mathbf{k} + \mathbf{0} - \mathbf{j}$
$= -2\mathbf{j} - 2\mathbf{k}$

2) h)

$(2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}) \times (3\mathbf{i} + \mathbf{j} - \mathbf{k})$
$= 2\mathbf{i} \times 3\mathbf{i} + 2\mathbf{i} \times \mathbf{j} - 2\mathbf{i} \times \mathbf{k} - 3\mathbf{j} \times 3\mathbf{i} - 3\mathbf{j} \times \mathbf{j} + 3\mathbf{j} \times \mathbf{k} + 5\mathbf{k} \times 3\mathbf{i} + 5\mathbf{k} \times \mathbf{j} - 5\mathbf{k} \times \mathbf{k}$
$= \mathbf{0} + 2\mathbf{k} + 2\mathbf{j} + 9\mathbf{k} - \mathbf{0} + 3\mathbf{i} + 15\mathbf{j} - 5\mathbf{i} - 0$
$= -2\mathbf{i} + 17\mathbf{j} + 11\mathbf{k}$

2) i)

$(3\mathbf{j} - 5\mathbf{i} - 3\mathbf{k}) \times (6\mathbf{i} - 8\mathbf{k})$
$= 18\mathbf{j} \times \mathbf{i} - 24\mathbf{j} \times \mathbf{k} - 30\mathbf{i} \times \mathbf{i} + 40\mathbf{i} \times \mathbf{k} - 18\mathbf{k} \times \mathbf{i} + 24\mathbf{k} \times \mathbf{k}$
$= -18\mathbf{k} - 24\mathbf{i} - \mathbf{0} - 40\mathbf{j} - 18\mathbf{j} + \mathbf{0}$
$= -24\mathbf{i} - 58\mathbf{j} - 18\mathbf{k}$

3) a)

$$\begin{pmatrix}1\\4\\2\end{pmatrix}\cdot\begin{pmatrix}8\\-5\\6\end{pmatrix}=1\times 8+4\times(-5)+2\times 6=0$$

$$\begin{pmatrix}3\\6\\1\end{pmatrix}\cdot\begin{pmatrix}8\\-5\\6\end{pmatrix}=3\times 8+6\times(-5)+1\times 6=0$$

$\therefore \begin{pmatrix}8\\-5\\6\end{pmatrix}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

3) b)

$$\begin{pmatrix}1\\4\\2\end{pmatrix}\times\begin{pmatrix}3\\6\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&4&2\\3&6&1\end{vmatrix}=(4\times 1-2\times 6)\mathbf{i}-(1\times 1-2\times 3)\mathbf{j}+(1\times 6-4\times 3)\mathbf{k}$$

$$=-8\mathbf{i}+5\mathbf{j}-6\mathbf{k}=\begin{pmatrix}-8\\5\\-6\end{pmatrix}$$

4) a)

$$\begin{pmatrix}2\\2\\3\end{pmatrix}\times\begin{pmatrix}3\\2\\2\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&2&3\\3&2&2\end{vmatrix}=(2\times 2-3\times 2)\mathbf{i}-(2\times 2-3\times 3)\mathbf{j}+(2\times 2-2\times 3)\mathbf{k}$$

$$=-2\mathbf{i}+5\mathbf{j}-2\mathbf{k}=\begin{pmatrix}-2\\5\\-2\end{pmatrix}$$

4) b)

$\mathbf{a}$ and $\mathbf{b}$ are both perpendicular to $\mathbf{a}\times\mathbf{b}$

5) a)

$$\begin{pmatrix}1\\2\\1\end{pmatrix}\times\begin{pmatrix}3\\1\\2\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&2&1\\3&1&2\end{vmatrix}=(2\times 2-1\times 1)\mathbf{i}-(1\times 2-1\times 3)\mathbf{j}+(1\times 1-2\times 3)\mathbf{k}$$

$$=3\mathbf{i}+\mathbf{j}-5\mathbf{k}=\begin{pmatrix}3\\1\\-5\end{pmatrix}$$

5) b)

$$\begin{pmatrix}5\\3\\2\end{pmatrix}\times\begin{pmatrix}4\\1\\6\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\5&3&2\\4&1&6\end{vmatrix}=(3\times6-2\times1)\mathbf{i}-(5\times6-2\times4)\mathbf{j}+(5\times1-3\times4)\mathbf{k}$$

$$=16\mathbf{i}-22\mathbf{j}-7\mathbf{k}=\begin{pmatrix}16\\-22\\-7\end{pmatrix}$$

5) c)

$$\begin{pmatrix}-2\\3\\-9\end{pmatrix}\times\begin{pmatrix}-10\\0\\7\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-2&3&-9\\-10&0&7\end{vmatrix}=(3\times7-(-9)\times0)\mathbf{i}-\big((-2)\times7-$$

$$(-9)\times(-10)\big)\mathbf{j}+\big((-2)\times0-3\times(-10)\big)\mathbf{k}$$

$$=21\mathbf{i}+104\mathbf{j}+30\mathbf{k}=\begin{pmatrix}21\\104\\30\end{pmatrix}$$

5) d)

$$\begin{pmatrix}3\\8\\-2\end{pmatrix}\times\begin{pmatrix}5\\-1\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&8&-2\\5&-1&1\end{vmatrix}=\big(8\times1-(-2)\times(-1)\big)\mathbf{i}-(3\times1-(-2)\times5)\mathbf{j}+$$

$$(3\times(-1)-8\times5)\mathbf{k}$$

$$=6\mathbf{i}-13\mathbf{j}-43\mathbf{k}$$

5) e)

$$\begin{pmatrix}-10\\2\\-3\end{pmatrix}\times\begin{pmatrix}-1\\5\\4\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-10&2&-3\\-1&5&4\end{vmatrix}=(2\times4-(-3)\times5)\mathbf{i}-\big((-10)\times4-$$

$$(-3)\times(-1)\big)\mathbf{j}+\big((-10)\times5-2\times(-1)\big)\mathbf{k}$$

$$=23\mathbf{i}+43\mathbf{j}-48\mathbf{k}$$

5) f)

$$\begin{pmatrix}3\\0\\7\end{pmatrix}\times\begin{pmatrix}0\\-2\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&0&7\\0&-2&-1\end{vmatrix}=\big(0\times(-1)-7\times(-2)\big)\mathbf{i}-(3\times(-1)-7\times0)\mathbf{j}+$$

$$(3\times(-2)-0\times0)\mathbf{k}$$

$$=14\mathbf{i}+3\mathbf{j}-6\mathbf{k}$$

5) g)

$$\begin{pmatrix}5\\0\\0\end{pmatrix}\times\begin{pmatrix}2\\-3\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\5&0&0\\2&-3&1\end{vmatrix}=\big(0\times1-0\times(-3)\big)\mathbf{i}-(5\times1-0\times2)\mathbf{j}+$$
$$(5\times(-3)-0\times2)\mathbf{k}$$
$$=-5\mathbf{j}-15\mathbf{k}$$

5) h)

$$\begin{pmatrix}-3\\0\\10\end{pmatrix}\times\begin{pmatrix}0\\3\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-3&0&10\\0&3&0\end{vmatrix}=(0\times0-10\times3)\mathbf{i}-\big((-3)\times0-10\times0\big)\mathbf{j}+$$
$$\big((-3)\times3-0\times0\big)\mathbf{k}$$
$$=-30\mathbf{i}-9\mathbf{k}$$

6)

$$\begin{pmatrix}2\\0\\-3\end{pmatrix}\times\begin{pmatrix}0\\-3\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&0&-3\\0&-3&-1\end{vmatrix}=\big(0\times(-1)-(-3)\times(-3)\big)\mathbf{i}-(2\times(-1)-$$
$$(-3)\times0)\mathbf{j}+(2\times(-3)-0\times0)\mathbf{k}$$
$$=-9\mathbf{i}+2\mathbf{j}-6\mathbf{k}$$
$$|-9\mathbf{i}+2\mathbf{j}-6\mathbf{k}|=\sqrt{(-9)^2+2^2+(-6)^2}=\sqrt{121}=11$$

So the unit vector is $\frac{1}{11}(-9\mathbf{i}+2\mathbf{j}-6\mathbf{k})=-\frac{9}{11}\mathbf{i}+\frac{2}{11}\mathbf{j}-\frac{6}{11}\mathbf{k}$

Or $\begin{pmatrix}-\frac{9}{11}\\\frac{2}{11}\\-\frac{6}{11}\end{pmatrix}$

7)

$\begin{pmatrix}-1\\1\\2\end{pmatrix}\times\begin{pmatrix}5\\-2\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-1&1&2\\5&-2&0\end{vmatrix}=\big(1\times0-2\times(-2)\big)\mathbf{i}-\big((-1)\times0-2\times5\big)\mathbf{j}+\big((-1)\times(-2)-1\times5\big)\mathbf{k}$

$=4\mathbf{i}+10\mathbf{j}-3\mathbf{k}$

$|4\mathbf{i}+10\mathbf{j}-3\mathbf{k}|=\sqrt{4^2+10^2+(-3)^2}=\sqrt{125}=5\sqrt{5}$

So the unit vector is $\frac{1}{5\sqrt{5}}(4\mathbf{i}+10\mathbf{j}-3\mathbf{k})=\frac{4\sqrt{5}}{25}\mathbf{i}+\frac{2\sqrt{5}}{5}\mathbf{j}-\frac{3\sqrt{5}}{25}\mathbf{k}$

Or $\begin{pmatrix}\frac{4\sqrt{5}}{25}\\\frac{2\sqrt{5}}{5}\\-\frac{3\sqrt{5}}{25}\end{pmatrix}$

8)

$\begin{pmatrix}1\\a\\2\end{pmatrix}\times\begin{pmatrix}a\\2\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&a&2\\a&2&3\end{vmatrix}=(a\times3-2\times2)\mathbf{i}-(1\times3-2\times a)\mathbf{j}+(1\times2-a\times a)\mathbf{k}$

$=(3a-4)\mathbf{i}+(2a-3)\mathbf{j}+(2-a^2)\mathbf{k}$

Or $\begin{pmatrix}3a-4\\2a-3\\2-a^2\end{pmatrix}$

9)

$\begin{pmatrix}p\\3\\-2\end{pmatrix}\times\begin{pmatrix}1-p\\2\\p\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\p&3&-2\\1-p&2&p\end{vmatrix}=(3\times p-(-2)\times2)\mathbf{i}-\big(p\times p-(-2)\times(1-p)\big)\mathbf{j}+\big(p\times2-3\times(1-p)\big)\mathbf{k}$

$=(3p+4)\mathbf{i}+(-p^2+2p-2)\mathbf{j}+(5p-3)\mathbf{k}$

Or $\begin{pmatrix}3p+4\\-p^2+2p-2\\5p-3\end{pmatrix}$

10)

$$\begin{pmatrix} 1 \\ q \\ 1-q \end{pmatrix} \times \begin{pmatrix} q-1 \\ 1 \\ q \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & q & 1-q \\ q-1 & 1 & q \end{vmatrix} = (q \times q - (1-q) \times 1)\mathbf{i} - \big(1 \times q - (1-q) \times (q-1)\big)\mathbf{j} + \big(1 \times 1 - q \times (q-1)\big)\mathbf{k}$$

$$= (q^2 + q - 1)\mathbf{i} + (-q^2 + q - 1)\mathbf{j} + (-q^2 + q + 1)\mathbf{k}$$

Or $\begin{pmatrix} q^2 + q - 1 \\ -q^2 + q - 1 \\ -q^2 + q + 1 \end{pmatrix}$

11) a)

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 2 \\ -7 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ -8 \\ -11 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -7 & 3 \\ 4 & -8 & -11 \end{vmatrix} = \big((-7) \times (-11) - 3 \times (-8)\big)\mathbf{i} - (2 \times (-11) - 3 \times 4)\mathbf{j} + (2 \times (-8) - (-7) \times 4)\mathbf{k}$$

$$= 101\mathbf{i} + 34\mathbf{j} + 12\mathbf{k} = \begin{pmatrix} 101 \\ 34 \\ 12 \end{pmatrix}$$

11) b)

$$\mathbf{b} \times \mathbf{a} = \begin{pmatrix} 4 \\ -8 \\ -11 \end{pmatrix} \times \begin{pmatrix} 2 \\ -7 \\ 3 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -8 & -11 \\ 2 & -7 & 3 \end{vmatrix} = \big((-8) \times 3 - (-11) \times (-7)\big)\mathbf{i} - (4 \times 3 - (-11) \times 2)\mathbf{j} + (4 \times (-7) - (-8) \times 2)\mathbf{k}$$

$$= -101\mathbf{i} - 34\mathbf{j} - 12\mathbf{k} = \begin{pmatrix} -101 \\ -34 \\ -12 \end{pmatrix}$$

11) c)

As $\mathbf{a} \times \mathbf{b} = 101\mathbf{i} + 34\mathbf{j} + 12\mathbf{k}$

and $\mathbf{b} \times \mathbf{a} = -101\mathbf{i} - 34\mathbf{j} - 12\mathbf{k}$

we have that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$

12)

$\begin{pmatrix}3\\-6\\12\end{pmatrix}\times\begin{pmatrix}-4\\8\\-16\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&-6&12\\-4&8&-16\end{vmatrix}=\big((-6)\times(-16)-12\times 8\big)\mathbf{i}-\big(3\times(-16)-12\times(-4)\big)\mathbf{j}+\big(3\times 8-(-6)\times(-4)\big)\mathbf{k}$

$=0\mathbf{i}+0\mathbf{j}+0\mathbf{k}=\mathbf{0}$

$\therefore$ the vectors are parallel

13)

Area of Triangle $=\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|=\frac{1}{2}\left|\sqrt{2}\mathbf{i}-9\mathbf{j}-\sqrt{5}\mathbf{k}\right|$

$=\frac{1}{2}\sqrt{\left(\sqrt{2}\right)^2+(-9)^2+\left(-\sqrt{5}\right)^2}=\frac{1}{2}\sqrt{2+81+5}=\frac{1}{2}\sqrt{88}=\frac{1}{2}\times 2\sqrt{22}=\sqrt{22}$ units^2

14)

$\begin{pmatrix}3\\-2\\5\end{pmatrix}\times\begin{pmatrix}8\\3\\5\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&-2&5\\8&3&5\end{vmatrix}=\big((-2)\times 5-5\times 3\big)\mathbf{i}-(3\times 5-5\times 8)\mathbf{j}+(3\times 3-(-2)\times 8)\mathbf{k}$

$=-25\mathbf{i}+25\mathbf{j}+25\mathbf{k}$

Area of Parallelogram $=|-25\mathbf{i}+25\mathbf{j}+25\mathbf{k}|=\sqrt{(-25)^2+25^2+25^2}$

$=\sqrt{1875}=25\sqrt{3}$ units^2

15) a)

$\overrightarrow{OA}=\overrightarrow{BC}=\begin{pmatrix}3\\6\\12\end{pmatrix}$ and $\overrightarrow{OB}=\overrightarrow{AC}=\begin{pmatrix}5\\-10\\20\end{pmatrix}$

$\overrightarrow{OC}=\overrightarrow{OA}+\overrightarrow{AC}=\overrightarrow{OA}+\overrightarrow{OB}=\begin{pmatrix}3\\6\\12\end{pmatrix}+\begin{pmatrix}5\\-10\\20\end{pmatrix}=\begin{pmatrix}8\\-4\\32\end{pmatrix}$

$\therefore C(8,-4,32)$

15) b)

$\begin{pmatrix}3\\6\\12\end{pmatrix}\times\begin{pmatrix}5\\-10\\20\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&6&12\\5&-10&20\end{vmatrix}=\big(6\times20-12\times(-10)\big)\mathbf{i}-(3\times20-12\times5)\mathbf{j}+$
$(3\times(-10)-6\times5)\mathbf{k}$
$=240\mathbf{i}+0\mathbf{j}-60\mathbf{k}$
Area of Parallelogram $=|240\mathbf{i}-60\mathbf{k}|=\sqrt{240^2+(-60)^2}$
$=\sqrt{61200}=60\sqrt{17}$ units^2

16)

$\overrightarrow{OA}=\begin{pmatrix}3\\-6\\-2\end{pmatrix}$ and $\overrightarrow{OB}=\begin{pmatrix}4\\-6\\0\end{pmatrix}$
$\begin{pmatrix}3\\-6\\-2\end{pmatrix}\times\begin{pmatrix}4\\-6\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&-6&-2\\4&-6&0\end{vmatrix}=\big((-6)\times0-(-2)\times(-6)\big)\mathbf{i}-(3\times0-(-2)\times4)\mathbf{j}+$
$(3\times(-6)-(-6)\times4)\mathbf{k}$
$=-12\mathbf{i}-8\mathbf{j}+6\mathbf{k}$
Area of Triangle $=\frac{1}{2}|-12\mathbf{i}-8\mathbf{j}+6\mathbf{k}|=\frac{1}{2}\sqrt{(-12)^2+(-8)^2+6^2}$
$=\frac{1}{2}\sqrt{244}=\sqrt{61}$ units^2

17) a)

Let A, B and C be $(-1,0,1)$, $(2,1,0)$ and $(1,1,-1)$ respectively.
$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}2\\1\\0\end{pmatrix}-\begin{pmatrix}-1\\0\\1\end{pmatrix}=\begin{pmatrix}3\\1\\-1\end{pmatrix}$
$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=\begin{pmatrix}1\\1\\-1\end{pmatrix}-\begin{pmatrix}-1\\0\\1\end{pmatrix}=\begin{pmatrix}2\\1\\-2\end{pmatrix}$
$\begin{pmatrix}3\\1\\-1\end{pmatrix}\times\begin{pmatrix}2\\1\\-2\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&1&-1\\2&1&-2\end{vmatrix}=(1\times(-2)-(-1)\times1)\mathbf{i}-(3\times(-2)-(-1)\times2)\mathbf{j}+$
$(3\times1-1\times2)\mathbf{k}$
$=-\mathbf{i}-4\mathbf{j}+\mathbf{k}$
Area of Triangle $=\frac{1}{2}|-\mathbf{i}-4\mathbf{j}+\mathbf{k}|=\frac{1}{2}\sqrt{(-1)^2+(-4)^2+1^2}$
$=\frac{1}{2}\sqrt{18}=\frac{3\sqrt{2}}{2}$ units^2

17) b)

Let A, B and C be $(3,-8,1)$, $(2,1,-3)$ and $(1,2,3)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} - \begin{pmatrix} 3 \\ -8 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 9 \\ -4 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ -8 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 10 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 9 \\ -4 \end{pmatrix} \times \begin{pmatrix} -2 \\ 10 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 9 & -4 \\ -2 & 10 & 2 \end{vmatrix} = (9 \times 2 - (-4) \times 10)\mathbf{i} - \big((-1) \times 2 -$$

$$(-4) \times (-2)\big)\mathbf{j} + \big((-1) \times 10 - 9 \times (-2)\big)\mathbf{k}$$

$$= 58\mathbf{i} - 10\mathbf{j} + 8\mathbf{k}$$

Area of Triangle $= \frac{1}{2}|58\mathbf{i} - 10\mathbf{j} + 8\mathbf{k}| = \frac{1}{2}\sqrt{58^2 + (-10)^2 + 8^2}$

$= \frac{1}{2}\sqrt{3528} = 21\sqrt{2}$ units^2

17) c)

Let A, B and C be $(4,2,-6)$, $(3,0,-1)$ and $(8,3,-5)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \\ -6 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 5 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 8 \\ 3 \\ -5 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \\ -6 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ -2 \\ 5 \end{pmatrix} \times \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 5 \\ 4 & 1 & 1 \end{vmatrix} = \big((-2) \times 1 - 5 \times 1\big)\mathbf{i} - \big((-1) \times 1 - 5 \times 4\big)\mathbf{j} +$$

$$\big((-1) \times 1 - (-2) \times 4\big)\mathbf{k}$$

$$= -7\mathbf{i} + 21\mathbf{j} + 7\mathbf{k}$$

Area of Triangle $= \frac{1}{2}|-7\mathbf{i} + 21\mathbf{j} + 7\mathbf{k}| = \frac{1}{2}\sqrt{(-7)^2 + 21^2 + 7^2}$

$= \frac{1}{2}\sqrt{539} = \frac{7}{2}\sqrt{11}$ units^2

17) d)

Let A, B and C be $(-1, 4, 5)$, $(9, -2, -3)$ and $(0, 5, 5)$ respectively.

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 9 \\ -2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 10 \\ -6 \\ -8 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 5 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 10 \\ -6 \\ -8 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 10 & -6 & -8 \\ 1 & 1 & 0 \end{vmatrix} = \big((-6) \times 0 - (-8) \times 1\big)\mathbf{i} - (10 \times 0 - (-8) \times 1)\mathbf{j} +$

$(10 \times 1 - (-6) \times 1)\mathbf{k}$

$= 8\mathbf{i} - 8\mathbf{j} + 16\mathbf{k}$

Area of Triangle $= \frac{1}{2}|8\mathbf{i} - 8\mathbf{j} + 16\mathbf{k}| = \frac{1}{2}\sqrt{8^2 + (-8)^2 + 16^2}$

$= \frac{1}{2}\sqrt{384} = 4\sqrt{6}$ units^2

18) a)

$\overrightarrow{OA} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 9 \\ -4 \\ 2 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} -17 \\ 16 \\ -8 \end{pmatrix}$

$p\begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} + q\begin{pmatrix} 9 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} -17 \\ 16 \\ -8 \end{pmatrix}$

$5p + 9q = -17$ (1)

$2p - 4q = 16$ (2)

$\therefore p = 2, q = -3$

Check:

$2(-1) + (-3)(2) = -8$

$\therefore 2\overrightarrow{OA} - 3\overrightarrow{OB} = \overrightarrow{OC}$

So $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are linearly dependent.

18) b)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 9 \\ -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 3 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -17 \\ 16 \\ -8 \end{pmatrix} - \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -22 \\ 14 \\ -7 \end{pmatrix}$

$\begin{pmatrix} 4 \\ -6 \\ 3 \end{pmatrix} \times \begin{pmatrix} -22 \\ 14 \\ -7 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -6 & 3 \\ -22 & 14 & -7 \end{vmatrix} = \big((-6) \times (-7) - 3 \times 14\big)\mathbf{i} - \big(4 \times (-7) - 3 \times (-22)\big)\mathbf{j} + \big(4 \times 14 - (-6) \times (-22)\big)\mathbf{k}$

$= 0\mathbf{i} - 38\mathbf{j} - 76\mathbf{k}$

Area of Triangle $= \frac{1}{2}|-38\mathbf{j} - 76\mathbf{k}| = \frac{1}{2}\sqrt{(-38)^2 + (-76)^2}$

$= \frac{1}{2}\sqrt{7220} = 19\sqrt{5}$ units^2

19)

$\overrightarrow{OA} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 4 \\ -3 \\ -1 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} -22 \\ 6 \\ 24 \end{pmatrix}$

$p\begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + q\begin{pmatrix} 4 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -22 \\ 6 \\ 24 \end{pmatrix}$

$2p + 4q = -22$ (1)

$-5p - 3q = 6$ (2)

$\therefore p = 3, q = -7$

Check:

$6(3) + (-1)(-7) = 25 \neq 24$

So $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are linearly independent.

20) a)

$\begin{pmatrix}2\\3\\-1\end{pmatrix}\times\begin{pmatrix}4\\2\\-3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&-1\\4&2&-3\end{vmatrix}=(3\times(-3)-(-1)\times 2)\mathbf{i}-(2\times(-3)-(-1)\times 4)\mathbf{j}+(2\times 2-3\times 4)\mathbf{k}$

$=-7\mathbf{i}+2\mathbf{j}-8\mathbf{k}$

$\begin{pmatrix}-7\\2\\-8\end{pmatrix}\cdot\begin{pmatrix}-2\\1\\2\end{pmatrix}=(-7)\times(-2)+2\times 1+(-8)\times 2=0$

$\therefore\begin{pmatrix}2\\3\\-1\end{pmatrix},\begin{pmatrix}4\\2\\-3\end{pmatrix}$ and $\begin{pmatrix}-2\\1\\2\end{pmatrix}$ are coplanar.

20) b)

$\begin{pmatrix}-4\\3\\6\end{pmatrix}\times\begin{pmatrix}-5\\3\\-8\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&3&6\\-5&3&-8\end{vmatrix}=(3\times(-8)-6\times 3)\mathbf{i}-\big((-4)\times(-8)-6\times(-5)\big)\mathbf{j}+\big((-4)\times 3-3\times(-5)\big)\mathbf{k}$

$=-42\mathbf{i}-62\mathbf{j}+3\mathbf{k}$

$\begin{pmatrix}-42\\-62\\3\end{pmatrix}\cdot\begin{pmatrix}3\\-9\\-144\end{pmatrix}=(-42)\times 3+(-62)\times(-9)+3\times(-144)=0$

$\therefore\begin{pmatrix}-4\\3\\6\end{pmatrix},\begin{pmatrix}-5\\3\\-8\end{pmatrix}$ and $\begin{pmatrix}3\\-9\\-144\end{pmatrix}$ are coplanar.

21) a)

$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}-4\\8\\1\end{pmatrix}\times\begin{pmatrix}4\\-6\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&8&1\\4&-6&3\end{vmatrix}=\big(8\times 3-1\times(-6)\big)\mathbf{i}-\big((-4)\times 3-1\times 4\big)\mathbf{j}+\big((-4)\times(-6)-8\times 4\big)\mathbf{k}$

$=30\mathbf{i}+16\mathbf{j}-8\mathbf{k}=\begin{pmatrix}30\\16\\-8\end{pmatrix}$

21) b)

$$\sin\theta = \frac{\left|\begin{pmatrix}-4\\8\\1\end{pmatrix}\times\begin{pmatrix}4\\-6\\3\end{pmatrix}\right|}{\left|\begin{pmatrix}-4\\8\\1\end{pmatrix}\right|\left|\begin{pmatrix}4\\-6\\3\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}30\\16\\-8\end{pmatrix}\right|}{\left|\begin{pmatrix}-4\\8\\1\end{pmatrix}\right|\left|\begin{pmatrix}4\\-6\\3\end{pmatrix}\right|}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{30^2+16^2+(-8)^2}}{\sqrt{(-4)^2+8^2+1^2}\times\sqrt{4^2+(-6)^2+3^2}}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{1220}}{\sqrt{81}\times\sqrt{61}} = \frac{2\sqrt{305}}{9\sqrt{61}} = \frac{2\sqrt{5}}{9}$$

22) a)

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix}1\\0\\1\end{pmatrix}\times\begin{pmatrix}2\\-1\\0\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 0 & 1\\2 & -1 & 0\end{vmatrix} = \big(0\times 0 - 1\times(-1)\big)\mathbf{i} - (1\times 0 - 1\times 2)\mathbf{j} +$$

$$(1\times(-1) - 0\times 2)\mathbf{k}$$

$$= \mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

$$\sin\theta = \frac{\left|\begin{pmatrix}1\\0\\1\end{pmatrix}\times\begin{pmatrix}2\\-1\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\1\end{pmatrix}\right|\left|\begin{pmatrix}2\\-1\\0\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}1\\2\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\1\end{pmatrix}\right|\left|\begin{pmatrix}2\\-1\\0\end{pmatrix}\right|}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{1^2+2^2+(-1)^2}}{\sqrt{1^2+0^2+1^2}\times\sqrt{2^2+(-1)^2+0^2}}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{6}}{\sqrt{2}\times\sqrt{5}} = \frac{\sqrt{15}}{5} \Rightarrow \theta = 50.768\ldots°$$

Acute angle is 50.8° (1 DP)

22) b)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}-7\\12\\1\end{pmatrix}\times\begin{pmatrix}3\\-20\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-7&12&1\\3&-20&-1\end{vmatrix}=\big(12\times(-1)-1\times(-20)\big)\mathbf{i}-$$
$$\big((-7)\times(-1)-1\times3\big)\mathbf{j}+\big((-7)\times(-20)-12\times3\big)\mathbf{k}$$
$$=8\mathbf{i}+4\mathbf{j}+104\mathbf{k}$$
$$\sin\theta=\frac{\left|\begin{pmatrix}-7\\12\\1\end{pmatrix}\times\begin{pmatrix}3\\-20\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}-7\\12\\1\end{pmatrix}\right|\left|\begin{pmatrix}3\\-20\\-1\end{pmatrix}\right|}=\frac{\left|\begin{pmatrix}8\\4\\104\end{pmatrix}\right|}{\left|\begin{pmatrix}-7\\12\\1\end{pmatrix}\right|\left|\begin{pmatrix}3\\-20\\-1\end{pmatrix}\right|}$$
$$\Rightarrow\sin\theta=\frac{\sqrt{8^2+4^2+104^2}}{\sqrt{(-7)^2+12^2+1^2}\times\sqrt{3^2+(-20)^2+(-1)^2}}$$
$$\Rightarrow\sin\theta=\frac{\sqrt{10896}}{\sqrt{194}\times\sqrt{410}}=\sqrt{\frac{10896}{79540}}\Rightarrow\theta=21.722\ldots°$$

Acute angle is 21.7° (1 DP)

22) c)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}2\sqrt{2}\\0\\7\sqrt{2}\end{pmatrix}\times\begin{pmatrix}5\sqrt{2}\\2\sqrt{2}\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2\sqrt{2}&0&7\sqrt{2}\\5\sqrt{2}&2\sqrt{2}&0\end{vmatrix}=\big(0\times0-7\sqrt{2}\times2\sqrt{2}\big)\mathbf{i}-$$
$$\big(2\sqrt{2}\times0-7\sqrt{2}\times5\sqrt{2}\big)\mathbf{j}+\big(2\sqrt{2}\times2\sqrt{2}-0\times5\sqrt{2}\big)\mathbf{k}$$
$$=-28\mathbf{i}+70\mathbf{j}+8\mathbf{k}$$
$$\sin\theta=\frac{\left|\begin{pmatrix}2\sqrt{2}\\0\\7\sqrt{2}\end{pmatrix}\times\begin{pmatrix}5\sqrt{2}\\2\sqrt{2}\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}2\sqrt{2}\\0\\7\sqrt{2}\end{pmatrix}\right|\left|\begin{pmatrix}5\sqrt{2}\\2\sqrt{2}\\0\end{pmatrix}\right|}=\frac{\left|\begin{pmatrix}-28\\70\\8\end{pmatrix}\right|}{\left|\begin{pmatrix}2\sqrt{2}\\0\\7\sqrt{2}\end{pmatrix}\right|\left|\begin{pmatrix}5\sqrt{2}\\2\sqrt{2}\\0\end{pmatrix}\right|}$$
$$\Rightarrow\sin\theta=\frac{\sqrt{(-28)^2+70^2+8^2}}{\sqrt{\left(2\sqrt{2}\right)^2+0^2+\left(7\sqrt{2}\right)^2}\times\sqrt{\left(5\sqrt{2}\right)^2+\left(2\sqrt{2}\right)^2+0^2}}$$
$$\Rightarrow\sin\theta=\frac{\sqrt{5748}}{\sqrt{106}\times\sqrt{58}}=\sqrt{\frac{5748}{6148}}\Rightarrow\theta=75.222\ldots°$$

Acute angle is 75.2° (1 DP)

22) d)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}1\\0\\9\end{pmatrix}\times\begin{pmatrix}0\\-1\\-15\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&0&9\\0&-1&-15\end{vmatrix}=\big(0\times(-15)-9\times(-1)\big)\mathbf{i}-(1\times(-15)-9\times0)\mathbf{j}+(1\times(-1)-0\times0)\mathbf{k}$$

$$=9\mathbf{i}+15\mathbf{j}-\mathbf{k}$$

$$\sin\theta=\frac{\left|\begin{pmatrix}1\\0\\9\end{pmatrix}\times\begin{pmatrix}0\\-1\\-15\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\9\end{pmatrix}\right|\left|\begin{pmatrix}0\\-1\\-15\end{pmatrix}\right|}=\frac{\left|\begin{pmatrix}9\\15\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\9\end{pmatrix}\right|\left|\begin{pmatrix}0\\-1\\-15\end{pmatrix}\right|}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{9^2+15^2+(-1)^2}}{\sqrt{1^2+0^2+9^2}\times\sqrt{0^2+(-1)^2+(-15)^2}}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{307}}{\sqrt{82}\times\sqrt{226}}=\sqrt{\frac{307}{18532}}\Rightarrow\theta=7.394\ldots°$$

Acute angle is 7.4° (1 DP)

22) e)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}\sqrt{2}\\\sqrt{3}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{3}\\\sqrt{2}\\\sqrt{6}\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\\sqrt{2}&\sqrt{3}&0\\\sqrt{3}&\sqrt{2}&\sqrt{6}\end{vmatrix}=\big(\sqrt{3}\times\sqrt{6}-0\times\sqrt{2}\big)\mathbf{i}-\big(\sqrt{2}\times\sqrt{6}-0\times\sqrt{3}\big)\mathbf{j}+\big(\sqrt{2}\times\sqrt{2}-\sqrt{3}\times\sqrt{3}\big)\mathbf{k}$$

$$=3\sqrt{2}\mathbf{i}-2\sqrt{3}\mathbf{j}-\mathbf{k}$$

$$\sin\theta=\frac{\left|\begin{pmatrix}\sqrt{2}\\\sqrt{3}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{3}\\\sqrt{2}\\\sqrt{6}\end{pmatrix}\right|}{\left|\begin{pmatrix}\sqrt{2}\\\sqrt{3}\\0\end{pmatrix}\right|\left|\begin{pmatrix}\sqrt{3}\\\sqrt{2}\\\sqrt{6}\end{pmatrix}\right|}=\frac{\left|\begin{pmatrix}3\sqrt{2}\\-2\sqrt{3}\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}\sqrt{2}\\\sqrt{3}\\0\end{pmatrix}\right|\left|\begin{pmatrix}\sqrt{3}\\\sqrt{2}\\\sqrt{6}\end{pmatrix}\right|}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{\left(3\sqrt{2}\right)^2+\left(-2\sqrt{3}\right)^2+(-1)^2}}{\sqrt{\left(\sqrt{2}\right)^2+\left(\sqrt{3}\right)^2+0^2}\times\sqrt{\left(\sqrt{3}\right)^2+\left(\sqrt{2}\right)^2+\left(\sqrt{6}\right)^2}}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{31}}{\sqrt{5}\times\sqrt{11}}=\sqrt{\frac{31}{55}}\Rightarrow\theta=48.656\ldots°$$

Acute angle is 48.7° (1 DP)

22) f)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}\pi\\-\sqrt{2}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{5}\\1\\\pi\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\\pi&-\sqrt{2}&0\\\sqrt{5}&1&\pi\end{vmatrix}=\left(\left(-\sqrt{2}\right)\times\pi-0\times1\right)\mathbf{i}-\left(\pi\times\pi-0\times\sqrt{5}\right)\mathbf{j}+\left(\pi\times1-\left(-\sqrt{2}\right)\times\sqrt{5}\right)\mathbf{k}$$

$$=-\sqrt{2}\pi\mathbf{i}-\pi^2\mathbf{j}+\left(\pi+\sqrt{10}\right)\mathbf{k}$$

$$\sin\theta=\frac{\left|\begin{pmatrix}\pi\\-\sqrt{2}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{5}\\1\\\pi\end{pmatrix}\right|}{\left|\begin{pmatrix}\pi\\-\sqrt{2}\\0\end{pmatrix}\right|\left|\begin{pmatrix}\sqrt{5}\\1\\\pi\end{pmatrix}\right|}=\frac{\left|\begin{pmatrix}-\sqrt{2}\pi\\-\pi^2\\\pi+\sqrt{10}\end{pmatrix}\right|}{\left|\begin{pmatrix}\pi\\-\sqrt{2}\\0\end{pmatrix}\right|\left|\begin{pmatrix}\sqrt{5}\\1\\\pi\end{pmatrix}\right|}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{\left(-\sqrt{2}\pi\right)^2+(-\pi^2)^2+\left(\pi+\sqrt{10}\right)^2}}{\sqrt{\pi^2+\left(-\sqrt{2}\right)^2+0^2}\times\sqrt{\left(\sqrt{5}\right)^2+1^2+\pi^2}}$$

$$\Rightarrow\sin\theta=\frac{\sqrt{\pi^4+3\pi^2+2\sqrt{10}\pi+10}}{\sqrt{\pi^2+2}\times\sqrt{\pi^2+6}}=0.912625\ldots\Rightarrow\theta=65.870\ldots°$$

Acute angle is 65.9° (1 DP)

23)

$$\begin{pmatrix}a\\a\\5\end{pmatrix}\times\begin{pmatrix}-1\\2\\-4\end{pmatrix}=\begin{pmatrix}-22\\7\\9\end{pmatrix}\Rightarrow\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a&a&5\\-1&2&-4\end{vmatrix}=-22\mathbf{i}+7\mathbf{j}+9\mathbf{k}$$

$$\Rightarrow(a\times(-4)-5\times2)\mathbf{i}-\left(a\times(-4)-5\times(-1)\right)\mathbf{j}+\left(a\times2-a\times(-1)\right)\mathbf{k}$$

$$=-22\mathbf{i}+7\mathbf{j}+9\mathbf{k}$$

$$\Rightarrow(-4a-10)\mathbf{i}+(4a-5)\mathbf{j}+3a\mathbf{k}=-22\mathbf{i}+7\mathbf{j}+9\mathbf{k}$$

$$-4a-10=-22\Rightarrow a=3$$

$$4a-5=7\Rightarrow a=3$$

$$3a=9\Rightarrow a=3$$

24)

$\begin{pmatrix} p \\ 3 \\ -2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 16 \\ q \\ -2 \end{pmatrix} \Rightarrow \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ p & 3 & -2 \\ 4 & 5 & 2 \end{vmatrix} = 16\mathbf{i} + q\mathbf{j} - 2\mathbf{k}$

$\Rightarrow (3 \times 2 - (-2) \times 5)\mathbf{i} - (p \times 2 - (-2) \times 4)\mathbf{j} + (p \times 5 - 3 \times 4)\mathbf{k}$

$= 16\mathbf{i} + q\mathbf{j} - 2\mathbf{k}$

$\Rightarrow 16\mathbf{i} + (-2p - 8)\mathbf{j} + (5p - 12)\mathbf{k} = 16\mathbf{i} + q\mathbf{j} - 2\mathbf{k}$

$-2p - 8 = q$ (1)

$5p - 12 = -2 \Rightarrow p = 2$ (2)

Substitute (2) into (1)

$-2(2) - 8 = q \Rightarrow q = -12$

$\therefore p = 2$ and $q = -12$

25)

$\begin{pmatrix} 8 \\ 2m \\ -9 \end{pmatrix} \times \begin{pmatrix} 5 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -45 \\ -37 \\ 3n \end{pmatrix} \Rightarrow \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 8 & 2m & -9 \\ 5 & -3 & -1 \end{vmatrix} = -45\mathbf{i} - 37\mathbf{j} + 3n\mathbf{k}$

$\Rightarrow \big(2m \times (-1) - (-9) \times (-3)\big)\mathbf{i} - (8 \times (-1) - (-9) \times 5)\mathbf{j} + (8 \times (-3) - 2m \times 5)\mathbf{k} = -45\mathbf{i} - 37\mathbf{j} + 3n\mathbf{k}$

$\Rightarrow (-2m - 27)\mathbf{i} - 37\mathbf{j} + (-24 - 10m)\mathbf{k} = -45\mathbf{i} - 37\mathbf{j} + 3n\mathbf{k}$

$-2m - 27 = -45 \Rightarrow m = 9$

$-24 - 10 \times 9 = 3n \Rightarrow n = -38$

26)

$\begin{pmatrix} p \\ -3 \\ 8 \end{pmatrix} \times \begin{pmatrix} 10 \\ 5 \\ 2q \end{pmatrix} = \begin{pmatrix} -16 \\ 168 \\ 5r \end{pmatrix} \Rightarrow \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ p & -3 & 8 \\ 10 & 5 & 2q \end{vmatrix} = -16\mathbf{i} + 168\mathbf{j} + 5r\mathbf{k}$

$\Rightarrow \big((-3) \times 2q - 8 \times 5\big)\mathbf{i} - (p \times 2q - 8 \times 10)\mathbf{j} + (p \times 5 - (-3) \times 10)\mathbf{k} = -16\mathbf{i} + 168\mathbf{j} + 5r\mathbf{k}$

$\Rightarrow (-6q - 40)\mathbf{i} - (2pq - 80)\mathbf{j} + (5p + 30)\mathbf{k} = -16\mathbf{i} + 168\mathbf{j} + 5r\mathbf{k}$

$-6q - 40 = -16 \Rightarrow q = -4$

$-(2p \times (-4) - 80) = 168 \Rightarrow p = 11$

$5 \times 11 + 30 = 5r \Rightarrow r = 17$

27)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ q-1 & q & -3 \\ 2 & 2q-3 & q \end{vmatrix} = \big(q \times q - (-3) \times (2q-3)\big)\mathbf{i} - \big((q-1) \times q - (-3) \times 2\big)\mathbf{j} +$$

$$\big((q-1) \times (2q-3) - q \times 2\big)\mathbf{k}$$

$$= (q^2 + 6q - 9)\mathbf{i} - (q^2 - q + 6)\mathbf{j} + (2q^2 - 7q + 3)\mathbf{k}$$

If $\mathbf{a}$ is perpendicular to $\mathbf{b} \times \mathbf{c}$, then $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$

$$2 \times (q^2 + 6q - 9) + (-2) \times -(q^2 - q + 6) + (-1) \times (2q^2 - 7q + 3) = 0$$

$$\Rightarrow 2q^2 + 12q - 18 + 2q^2 - 2q + 12 - 2q^2 + 7q - 3 = 0$$

$$\Rightarrow 2q^2 + 17q - 9 = 0$$

$$\Rightarrow (2q-1)(q+9) = 0$$

$$\Rightarrow q = \tfrac{1}{2} \text{ or } q = -9$$

28) a)

$$(2\mathbf{a}) \times (5\mathbf{b}) = 10(\mathbf{a} \times \mathbf{b}) = 10\mathbf{c}$$

28) b)

$$\left(\tfrac{1}{2}\mathbf{b}\right) \times (8\mathbf{a}) = 4(\mathbf{b} \times \mathbf{a}) = -4(\mathbf{a} \times \mathbf{b}) = -4\mathbf{c}$$

29)

If $\mathbf{a}$ is parallel to both $\mathbf{b}$ and $\mathbf{c}$ then $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ and $\mathbf{a} \times \mathbf{c} = \mathbf{0}$

$$\therefore \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

30) a)

$$(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = \mathbf{a} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{a} - \mathbf{b} \times \mathbf{b}$$

$$= \mathbf{0} - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{b} - \mathbf{0} = -2\mathbf{a} \times \mathbf{b}$$

30) b)

$$(2\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + 3\mathbf{b}) = 2\mathbf{a} \times \mathbf{a} + 6\mathbf{a} \times \mathbf{b} - \mathbf{b} \times \mathbf{a} - 3\mathbf{b} \times \mathbf{b}$$

$$= 2 \times \mathbf{0} + 6\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{b} - 3 \times \mathbf{0} = 7\mathbf{a} \times \mathbf{b}$$

30) c)

$$(5\mathbf{b} - 6\mathbf{a}) \times (6\mathbf{b} - 5\mathbf{a}) = 30\mathbf{b} \times \mathbf{b} - 25\mathbf{b} \times \mathbf{a} - 36\mathbf{a} \times \mathbf{b} + 30\mathbf{a} \times \mathbf{a}$$

$$= 30 \times \mathbf{0} + 25\mathbf{a} \times \mathbf{b} - 36\mathbf{a} \times \mathbf{b} + 30 \times \mathbf{0} = -11\mathbf{a} \times \mathbf{b} = 11\mathbf{b} \times \mathbf{a}$$

31) a)

$(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{c})=\mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{c}$

$=\mathbf{0}-\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}=-\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}$

31) b)

$(\mathbf{c}-\mathbf{b})\times(\mathbf{b}-\mathbf{a})=\mathbf{c}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}-\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{a}$

$=-\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}-\mathbf{0}-\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}$

31) c)

$(\mathbf{a}-\mathbf{b}+\mathbf{c})\times(\mathbf{b}-\mathbf{a}+\mathbf{c})$

$=\mathbf{a}\times\mathbf{b}-\mathbf{a}\times\mathbf{a}+\mathbf{a}\times\mathbf{c}-\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}+\mathbf{c}\times\mathbf{c}$

$=\mathbf{a}\times\mathbf{b}-\mathbf{0}+\mathbf{a}\times\mathbf{c}-\mathbf{0}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}+\mathbf{0}$

$=2\mathbf{a}\times\mathbf{c}-2\mathbf{b}\times\mathbf{c}$

31) d)

$(2\mathbf{a}+\mathbf{b}-3\mathbf{c})\times(4\mathbf{a}-3\mathbf{b}+\mathbf{c})$

$=8\mathbf{a}\times\mathbf{a}-6\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+4\mathbf{b}\times\mathbf{a}-3\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}-12\mathbf{c}\times\mathbf{a}+9\mathbf{c}\times\mathbf{b}-3\mathbf{c}\times\mathbf{c}$

$=8\times\mathbf{0}-6\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}-4\mathbf{a}\times\mathbf{b}-3\times\mathbf{0}+\mathbf{b}\times\mathbf{c}+12\mathbf{a}\times\mathbf{c}-9\mathbf{b}\times\mathbf{c}-3\times\mathbf{0}$

$=-10\mathbf{a}\times\mathbf{b}+14\mathbf{a}\times\mathbf{c}-8\mathbf{b}\times\mathbf{c}$

32)

$(2\mathbf{a}-\mathbf{b})\times(6\mathbf{a}-5\mathbf{b})=\mathbf{0}$

$\Rightarrow 12\mathbf{a}\times\mathbf{a}-10\mathbf{a}\times\mathbf{b}-6\mathbf{b}\times\mathbf{a}+5\mathbf{b}\times\mathbf{b}=\mathbf{0}$

$\Rightarrow 12\times\mathbf{0}-10\mathbf{a}\times\mathbf{b}+6\mathbf{a}\times\mathbf{b}+5\times\mathbf{0}=\mathbf{0}$

$\Rightarrow -4\mathbf{a}\times\mathbf{b}=\mathbf{0}$

$\Rightarrow \mathbf{a}\times\mathbf{b}=\mathbf{0}$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are parallel

33) a)

$(\mathbf{a}+2\mathbf{b})\times(\mathbf{a}-\mathbf{c})$

$=\mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+2\mathbf{b}\times\mathbf{a}-2\mathbf{b}\times\mathbf{c}$

$=\mathbf{0}-\mathbf{a}\times\mathbf{c}-2\mathbf{a}\times\mathbf{b}-2\mathbf{b}\times\mathbf{c}$

$=-\mathbf{a}\times\mathbf{c}-2\mathbf{a}\times\mathbf{b}-2\mathbf{b}\times\mathbf{c}$

$=-\mathbf{j}-2\mathbf{k}-2\mathbf{i}$

$=-2\mathbf{i}-\mathbf{j}-2\mathbf{k}$

33) b)

$(2\mathbf{a}+\mathbf{b}+\mathbf{c})\times(\mathbf{b}+\mathbf{c})$
$=2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{b}+\mathbf{c}\times\mathbf{c}$
$=2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+\mathbf{0}+\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{c}+\mathbf{0}$
$=2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}$
$=2\mathbf{k}+2\mathbf{j}$
$=2\mathbf{j}+2\mathbf{k}$

33) c)

$(3\mathbf{a}-2\mathbf{b}-\mathbf{c})\times(\mathbf{c}-4\mathbf{b}-3\mathbf{a})$
$=3\mathbf{a}\times\mathbf{c}-12\mathbf{a}\times\mathbf{b}-9\mathbf{a}\times\mathbf{a}-2\mathbf{b}\times\mathbf{c}+8\mathbf{b}\times\mathbf{b}+6\mathbf{b}\times\mathbf{a}-\mathbf{c}\times\mathbf{c}+4\mathbf{c}\times\mathbf{b}+3\mathbf{c}\times\mathbf{a}$
$=3\mathbf{a}\times\mathbf{c}-12\mathbf{a}\times\mathbf{b}-9\times\mathbf{0}-2\mathbf{b}\times\mathbf{c}+8\times\mathbf{0}-6\mathbf{a}\times\mathbf{b}-\mathbf{0}-4\mathbf{b}\times\mathbf{c}-3\mathbf{a}\times\mathbf{c}$
$=-18\mathbf{a}\times\mathbf{b}-6\mathbf{b}\times\mathbf{c}$
$=-18\mathbf{k}-6\mathbf{i}$
$=-6\mathbf{i}-18\mathbf{k}$

34)

$(2\mathbf{a}-3\mathbf{b}+8\mathbf{c})\times(7\mathbf{a}+6\mathbf{b}-16\mathbf{c})$
$=14\mathbf{a}\times\mathbf{a}+12\mathbf{a}\times\mathbf{b}-32\mathbf{a}\times\mathbf{c}-21\mathbf{b}\times\mathbf{a}-18\mathbf{b}\times\mathbf{b}+48\mathbf{b}\times\mathbf{c}+56\mathbf{c}\times\mathbf{a}+48\mathbf{c}\times\mathbf{b}-128\mathbf{c}\times\mathbf{c}$
$=14\times\mathbf{0}+12\mathbf{a}\times\mathbf{b}-32\mathbf{a}\times\mathbf{c}+21\mathbf{a}\times\mathbf{b}-18\times\mathbf{0}+48\mathbf{b}\times\mathbf{c}-56\mathbf{a}\times\mathbf{c}-48\mathbf{b}\times\mathbf{c}-128\times\mathbf{0}$
$=33\mathbf{a}\times\mathbf{b}-88\mathbf{a}\times\mathbf{c}$
$=33\times3\mathbf{i}-88\times4\mathbf{j}$
$=99\mathbf{i}-352\mathbf{j}$

35)

$(\mathbf{c}-\mathbf{b}-4\mathbf{a})\times(4\mathbf{a}-6\mathbf{b}+9\mathbf{c})$
$=4\mathbf{c}\times\mathbf{a}-6\mathbf{c}\times\mathbf{b}+9\mathbf{c}\times\mathbf{c}-4\mathbf{b}\times\mathbf{a}+6\mathbf{b}\times\mathbf{b}-9\mathbf{b}\times\mathbf{c}-16\mathbf{a}\times\mathbf{a}+24\mathbf{a}\times\mathbf{b}-36\mathbf{a}\times\mathbf{c}$
$=-4\mathbf{a}\times\mathbf{c}+6\mathbf{b}\times\mathbf{c}+9\times\mathbf{0}+4\mathbf{a}\times\mathbf{b}+6\times\mathbf{0}-9\mathbf{b}\times\mathbf{c}-16\times\mathbf{0}+24\mathbf{a}\times\mathbf{b}-36\mathbf{a}\times\mathbf{c}$
$=28\mathbf{a}\times\mathbf{b}-3\mathbf{b}\times\mathbf{c}-40\mathbf{a}\times\mathbf{c}$
$=28\times2\mathbf{k}-3\times3\mathbf{j}-40\times5\mathbf{i}$
$=-200\mathbf{i}-9\mathbf{j}+56\mathbf{k}$

36)

$(-2\mathbf{a}-6\mathbf{b}+3\mathbf{c})\times 5\mathbf{a}$

$=-10\mathbf{a}\times\mathbf{a}-30\mathbf{b}\times\mathbf{a}+15\mathbf{c}\times\mathbf{a}$

$=-10\times\mathbf{0}+30\mathbf{a}\times\mathbf{b}-15\mathbf{a}\times\mathbf{c}$

$=30\begin{pmatrix}4\\1\\-3\end{pmatrix}-15\begin{pmatrix}8\\-3\\2\end{pmatrix}=\begin{pmatrix}0\\75\\-120\end{pmatrix}$

37)

$(3\mathbf{a}-\mathbf{b}+2\mathbf{c})\times(3\mathbf{a}-2\mathbf{b}-5\mathbf{c})$

$=9\mathbf{a}\times\mathbf{a}-6\mathbf{a}\times\mathbf{b}-15\mathbf{a}\times\mathbf{c}-3\mathbf{b}\times\mathbf{a}+2\mathbf{b}\times\mathbf{b}+5\mathbf{b}\times\mathbf{c}+6\mathbf{c}\times\mathbf{a}-4\mathbf{c}\times\mathbf{b}-10\mathbf{c}\times\mathbf{c}$

$=9\times\mathbf{0}-6\mathbf{a}\times\mathbf{b}-15\mathbf{a}\times\mathbf{c}+3\mathbf{a}\times\mathbf{b}+2\times\mathbf{0}+5\mathbf{b}\times\mathbf{c}-6\mathbf{a}\times\mathbf{c}+4\mathbf{b}\times\mathbf{c}-10\times\mathbf{0}$

$=-6\begin{pmatrix}3\\0\\2\end{pmatrix}-15\begin{pmatrix}-2\\-1\\4\end{pmatrix}+3\begin{pmatrix}3\\0\\2\end{pmatrix}+5\begin{pmatrix}-1\\1\\2\end{pmatrix}-6\begin{pmatrix}-2\\-1\\4\end{pmatrix}+4\begin{pmatrix}-1\\1\\2\end{pmatrix}=\begin{pmatrix}24\\30\\-72\end{pmatrix}$

38) a)

$|(\mathbf{a}+\mathbf{b})\times(2\mathbf{a}-\mathbf{b})|=|2\mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{b}+2\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{b}|$

$=|2\times\mathbf{0}-\mathbf{a}\times\mathbf{b}-2\mathbf{a}\times\mathbf{b}-\mathbf{0}|$

$=|-3\mathbf{a}\times\mathbf{b}|$

$=3|\mathbf{a}\times\mathbf{b}|$

$=3|\mathbf{a}||\mathbf{b}|\sin 90°$

$=3|\mathbf{a}||\mathbf{b}|$

38) b)

$|(3\mathbf{a}-2\mathbf{b})\times(4\mathbf{a}+\mathbf{b})|=|12\mathbf{a}\times\mathbf{a}+3\mathbf{a}\times\mathbf{b}-8\mathbf{b}\times\mathbf{a}-2\mathbf{b}\times\mathbf{b}|$

$=|12\times\mathbf{0}+3\mathbf{a}\times\mathbf{b}+8\mathbf{a}\times\mathbf{b}-2\times\mathbf{0}|$

$=|11\mathbf{a}\times\mathbf{b}|$

$=11|\mathbf{a}\times\mathbf{b}|$

$=11|\mathbf{a}||\mathbf{b}|\sin 90°$

$=11|\mathbf{a}||\mathbf{b}|$

38) c)

$|(6\mathbf{b}-\mathbf{a})\times(9\mathbf{a}-\mathbf{b})| = |54\mathbf{b}\times\mathbf{a}-6\mathbf{b}\times\mathbf{b}-9\mathbf{a}\times\mathbf{a}+\mathbf{a}\times\mathbf{b}|$

$= |-54\mathbf{a}\times\mathbf{b}-6\times\mathbf{0}-9\times\mathbf{0}+\mathbf{a}\times\mathbf{b}|$

$= |-53\mathbf{a}\times\mathbf{b}|$

$= 53|\mathbf{a}\times\mathbf{b}|$

$= 53|\mathbf{a}||\mathbf{b}|\sin 90°$

$= 53|\mathbf{a}||\mathbf{b}|$

39)

$\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\,\hat{\mathbf{n}}$

If $\theta = 0°$ then $\sin 0° = 0$

If $\theta = 180°$ then $\sin 180° = 0$

If a and b are parallel, then the angle between them will be zero, hence $\mathbf{a}\times\mathbf{b} = \mathbf{0}$

40)

$(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b}) = \mathbf{0}$

$\Rightarrow \mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{b} = \mathbf{0}$

$\Rightarrow \mathbf{0}-\mathbf{a}\times\mathbf{b}-\mathbf{a}\times\mathbf{b}-\mathbf{0} = \mathbf{0}$

$\Rightarrow -2\mathbf{a}\times\mathbf{b} = \mathbf{0}$

$\Rightarrow \mathbf{a}\times\mathbf{b} = \mathbf{0}$

So $\mathbf{a}$ and $\mathbf{b}$ are parallel, so the angle between the vectors is either $0°$ or $180°$

41)

$\mathbf{a}\times\mathbf{b} = \mathbf{b}\times\mathbf{c} \Rightarrow \mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c} = \mathbf{0} \Rightarrow \mathbf{a}\times\mathbf{b}+\mathbf{c}\times\mathbf{b} = \mathbf{0} \Rightarrow (\mathbf{a}+\mathbf{c})\times\mathbf{b} = 0$

So $(\mathbf{a}+\mathbf{c})$ must be parallel to b, hence $\mathbf{a}+\mathbf{c} = k\mathbf{b}$

42)

$|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$

$\Rightarrow 7\sqrt{2} = 4\times 3\sqrt{2}\sin\theta$

$\Rightarrow \sin\theta = \frac{7}{12}$

$\therefore \cos\theta = \frac{\sqrt{12^2-7^2}}{12} = \frac{\sqrt{95}}{12}$

$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$

$\Rightarrow \mathbf{a}\cdot\mathbf{b} = 4\times 3\sqrt{2}\times\frac{\sqrt{95}}{12} = \sqrt{190}$

43)

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$

$\Rightarrow 10\sqrt{66} = 25 \times 2\sqrt{3} \cos\theta$

$\Rightarrow \cos\theta = \frac{\sqrt{22}}{5}$

$\therefore \sin\theta = \frac{\sqrt{5^2 - \left(\sqrt{22}\right)^2}}{5} = \frac{\sqrt{3}}{5}$

$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$

$\Rightarrow |\mathbf{a} \times \mathbf{b}| = 25 \times 2\sqrt{3} \times \frac{\sqrt{3}}{5} = 30$

44)

$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$

$\Rightarrow p = ab \sin\theta \Rightarrow \sin\theta = \frac{p}{ab}$

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$

$\Rightarrow q = ab \cos\theta \Rightarrow \cos\theta = \frac{q}{ab}$

$\sin^2\theta + \cos^2\theta = 1 \Rightarrow \left(\frac{p}{ab}\right)^2 + \left(\frac{q}{ab}\right)^2 = 1 \Rightarrow \frac{p^2}{a^2b^2} + \frac{q^2}{a^2b^2} = 1$

$\Rightarrow p^2 + q^2 = a^2b^2$

45) a)

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -1 \\ 2 & -1 & 1 \end{vmatrix}$$

$= \big(3 \times 1 - (-1) \times (-1)\big)\mathbf{i} - (2 \times 1 - (-1) \times 2)\mathbf{j} + (2 \times (-1) - 3 \times 2)\mathbf{k}$

$= 2\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}$

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -4 \\ -8 \end{pmatrix} = 3 \times 2 + (-2) \times (-4) + 1 \times (-8)$$

$= 6 + 8 - 8 = 6$

45) b)

$$\mathbf{b}\times\mathbf{c}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 3 & -3 & 1\\ 4 & 1 & 4\end{vmatrix}$$

$$=\big((-3)\times 4-1\times 1\big)\mathbf{i}-(3\times 4-1\times 4)\mathbf{j}+(3\times 1-(-3)\times 4)\mathbf{k}$$

$$=-13\mathbf{i}-8\mathbf{j}+15\mathbf{k}$$

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=\begin{pmatrix}4\\16\\12\end{pmatrix}\cdot\begin{pmatrix}-13\\-8\\15\end{pmatrix}=4\times(-13)+16\times(-8)+12\times 15$$

$$=-52-128+180=0$$

46)

$\mathbf{b}\times\mathbf{c}$ finds a vector perpendicular to both $\mathbf{b}$ and $\mathbf{c}$. If the scalar product between $\mathbf{a}$ and this perpendicular vector is zero, then $\mathbf{a}$ is also perpendicular to this vector. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are all perpendicular to $\mathbf{b}\times\mathbf{c}$, then $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ must be coplanar.

47)

$$\mathbf{a}\times\mathbf{b}=(a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k})\times(b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k})$$

$$=a_1b_1\mathbf{i}\times\mathbf{i}+a_1b_2\mathbf{i}\times\mathbf{j}+a_1b_3\mathbf{i}\times\mathbf{k}+a_2b_1\mathbf{j}\times\mathbf{i}+a_2b_2\mathbf{j}\times\mathbf{j}+a_2b_3\mathbf{j}\times\mathbf{k}+a_3b_1\mathbf{k}\times\mathbf{i}+a_3b_2\mathbf{k}\times\mathbf{j}+a_3b_3\mathbf{k}\times\mathbf{k}$$

$$=a_1b_1\times\mathbf{0}+a_1b_2\mathbf{i}\times\mathbf{j}+a_1b_3\mathbf{i}\times\mathbf{k}-a_2b_1\mathbf{i}\times\mathbf{j}+a_2b_2\times\mathbf{0}+a_2b_3\mathbf{j}\times\mathbf{k}-a_3b_1\mathbf{i}\times\mathbf{k}-a_3b_2\mathbf{j}\times\mathbf{k}+a_3b_3\times\mathbf{0}$$

$$=a_1b_2\mathbf{k}-a_1b_3\mathbf{j}-a_2b_1\mathbf{k}+a_2b_3\mathbf{i}+a_3b_1\mathbf{j}-a_3b_2\mathbf{i}$$

$$=(a_2b_3-a_3b_2)\mathbf{i}+(a_3b_1-a_1b_3)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}$$

$$=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}$$

48)

$$\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}$$

$$\mathbf{b}\times\mathbf{a}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ b_1 & b_2 & b_3\\ a_1 & a_2 & a_3\end{vmatrix}=(a_3b_2-a_2b_3)\mathbf{i}-(a_3b_1-a_1b_3)\mathbf{j}+(a_2b_1-a_1b_2)\mathbf{k}$$

Then $-\mathbf{b}\times\mathbf{a}=-(a_3b_2-a_2b_3)\mathbf{i}+(a_3b_1-a_1b_3)\mathbf{j}-(a_2b_1-a_1b_2)\mathbf{k}$

$$=(-a_3b_2+a_2b_3)\mathbf{i}-(-a_3b_1+a_1b_3)\mathbf{j}+(-a_2b_1+a_1b_2)\mathbf{k}$$

$$=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}=\mathbf{a}\times\mathbf{b}$$

$\therefore\ \mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$

49)

$$(\mathbf{a}+\mathbf{b})\times\mathbf{c}=\left(\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}+\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}\right)\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}=\begin{pmatrix}a_1+b_1\\a_2+b_2\\a_3+b_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}$$

$$=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1+b_1 & a_2+b_2 & a_3+b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

$$=\big((a_2+b_2)c_3-(a_3+b_3)c_2\big)\mathbf{i}-\big((a_1+b_1)c_3-(a_3+b_3)c_1\big)\mathbf{j}+\big((a_1+b_1)c_2-(a_2+b_2)c_2\big)\mathbf{k}$$

$$=(a_2c_3+b_2c_3-a_3c_2-b_3c_2)\mathbf{i}-(a_1c_3+b_1c_3-a_3c_1-b_3c_1)\mathbf{j}+(a_1c_2+b_1c_2-a_2c_2-b_2c_2)\mathbf{k}$$

$$=(a_2c_3-a_3c_2+b_2c_3-b_3c_2)\mathbf{i}-(a_1c_3-a_3c_1+b_1c_3-b_3c_1)\mathbf{j}+(a_1c_2-a_2c_2+b_1c_2-b_2c_2)\mathbf{k}$$

$$=(a_2c_3-a_3c_2)\mathbf{i}-(a_1c_3-a_3c_1)\mathbf{j}+(a_1c_2-a_2c_2)\mathbf{k}+(b_2c_3-b_3c_2)\mathbf{i}-(b_1c_3-b_3c_1)\mathbf{j}+(b_1c_2-b_2c_2)\mathbf{k}$$

$$=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ c_1 & c_2 & c_3\end{vmatrix}+\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

$$=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}+\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}=\mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{c}$$